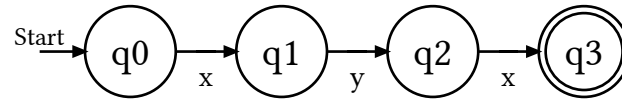


1) Four languages

Let $\Sigma = \{x, y\}$

1.a) $\{xyx \mid n > 0\}$

Since neither x or y rely on n we can draw a very simple **DFA**, therefore the language is regular:



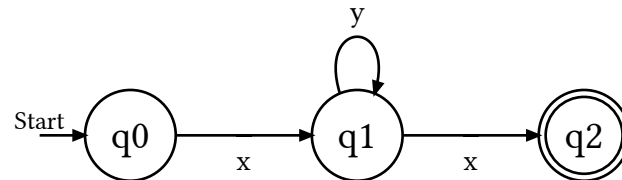
1.b) $\{x^n y x^n \mid n \geq 0\}$

This language requires knowledge about the first x group to guarantee the later x group is the same size, let's make a simple **CFG** for this non-regular language:

$$S \rightarrow xSx \mid y$$

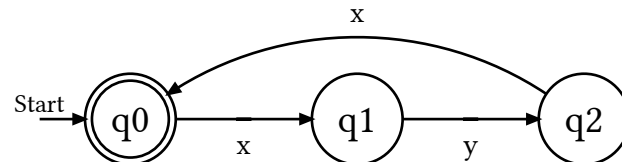
1.c) $\{xy^n x \mid n \geq 0\}$

This language is very clearly regular and we can use a slightly modified **DFA** from **a)** to show this:



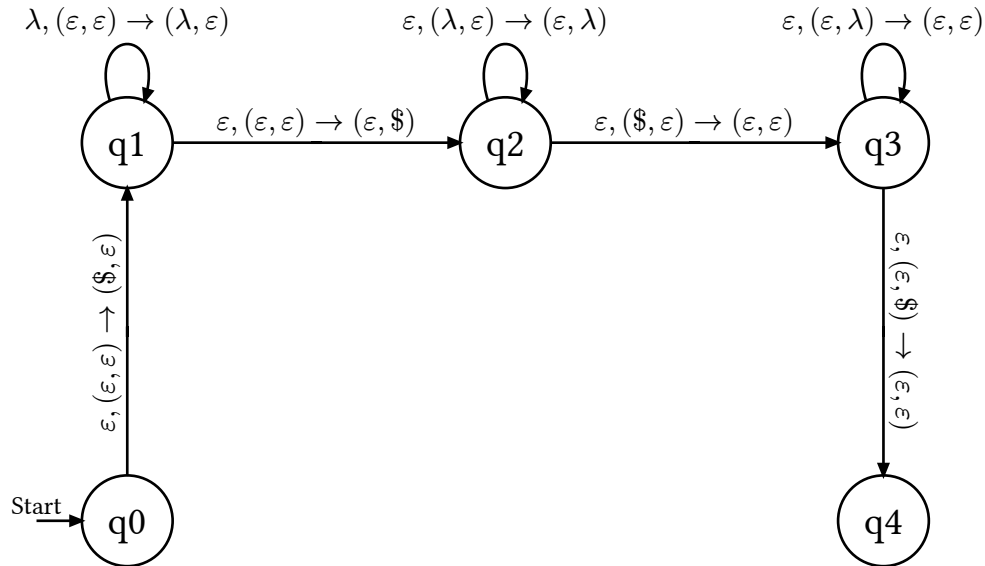
1.d) $\{(xyx)^n \mid n \geq 0\}$

This language is regular and we can use a **DFA** to show this:



2) Stacks on stacks

For ease of both reading and writing i'll be using λ to represent both a and b , since i won't ever be using them both in the same transition. The transitions will always be some version $a, (\varepsilon, \varepsilon) \rightarrow (a, \varepsilon)$ and never $a, (\varepsilon, \varepsilon) \rightarrow (b, \varepsilon)$



The transition $q_1 \rightarrow q_2$ is a *magic* transition that happens when we have finished the first half of tokens

3) Checksum continuation

Suppose that A is a CFL, then there exists a number p where, if S is any string in A of length p , then S may be divided into five pieces $S = uvxyz$ satisfying the following rules:

- i. $uv^i xy^i z \in A$ for each $i \geq 0$
- ii. $|vy| > 0$
- iii. $|vxy| \leq p$

3.a) pump it up (or down)

- We let p denote the pumping length
- We let $S = 0^p 1^i 1^{p-2i} 1^i 2^p 00$

According to pumping lemma S splits into $uvxyz$ so that i-iii. hold

$$u : 0^p, v : 1^i, x : 1^{p-2i}, y : 1^i, z : 2^p 00$$

- i. We offset the number of x 's by the number of both v 's and y 's so our string holds for each $i \geq 0$
- ii. $|vy|$ is $2 > 0$
- iii. We make $|vxy| = i + (p - 2i) + i = p$

Now we pump down and set $i = 0$ then $S' = 0^p 1^{p-2} 2^p 00 \notin A$ and since the checksum should now be 10 the language A is not context free